

IT Security 2

Unit 12 - Controlling Access to Network Resources with User-ID

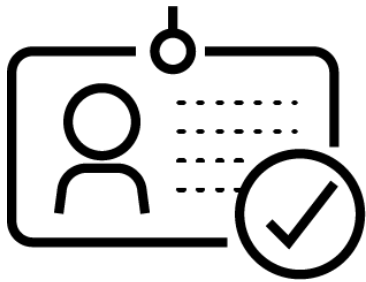
Instructor: Brian Krause

Announcements

Controlling Access to Network Resources with User-ID

Some slides were adapted from Palo-Alto for use in this class. Content is a mesh between vendor provided information and instructor comments.

CONTROLLING ACCESS TO NETWORK RESOURCES WITH USER-ID



KNOW THE WHO, CONTROL THE WHO

- User-ID overview
- User mapping methods overview
- Configure User-ID
- PAN-OS integrated agent configuration
- Configure group mapping
- User-ID and Security policy

Learning Objectives

After you complete this module, you should be able to:

- Identify the purpose and four main components of User-ID
- Identify available IP-to-username mapping methods
- Configure the PAN-OS[®] integrated agent to connect to monitored servers
- Configure username-to-group name mapping
- Implement User-ID in Security policy





User-ID overview

User mapping methods overview

Configure User-ID

PAN-OS integrated agent configuration

Configure group mapping

User-ID and Security policy

User-ID Purposes

- Identify users by username and user group.
- Create policies and display logs and reports based on usernames and group names.

Policies > Security

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	lab\lab users	any	Extranet	any	any	ssl web-browsing	application-default	Allow
2	Users_to_Internet	Users_Net	universal	Users_Net	any	lab\lab users	any	Internet	any	any	any	application-default	Allow
3	Danger-Simulated-Tr...	Danger	universal	Danger	any	lab\lab users	any	Danger	any	any	any	application-default	Allow

Monitor > Logs > Traffic

	RECEIVE TIME	TYPE	FROM ZONE	TO ZONE	SOURCE	SOURCE USER	DESTINATION	TO PORT	APPLICATION	ACTION	RULE
	07/22 22:39:12	end	Users_Net	Internet	192.168.1.254	lab\lab-user-id	107.178.249.217	443	paloalto-updates	allow	Users_to_Internet
	07/22 22:36:27	end	Users_Net	Internet	192.168.1.254	lab\lab-user-id	35.190.82.33	443	paloalto-updates	allow	Users_to_Internet
	07/22 22:34:06	end	Users_Net	Internet	192.168.1.254	lab\lab-user-id	107.178.249.217	443	paloalto-updates	allow	Users_to_Internet

User-ID Main Functions

Network Traffic

User Mapping:

Associate IP addresses with usernames.

IP Address

Username

LDAP

Group

Group Mapping: Learn group names and member users from LDAP.

**Policy Rules,
Logs, Reports**

For a Tech Doc about this topic, log into Live and search for "User-ID Overview"



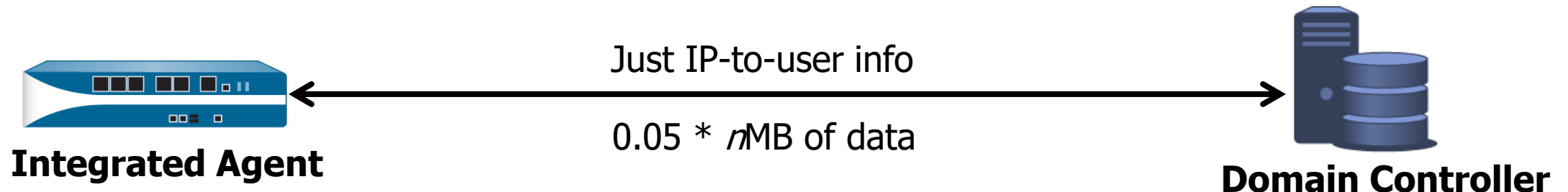
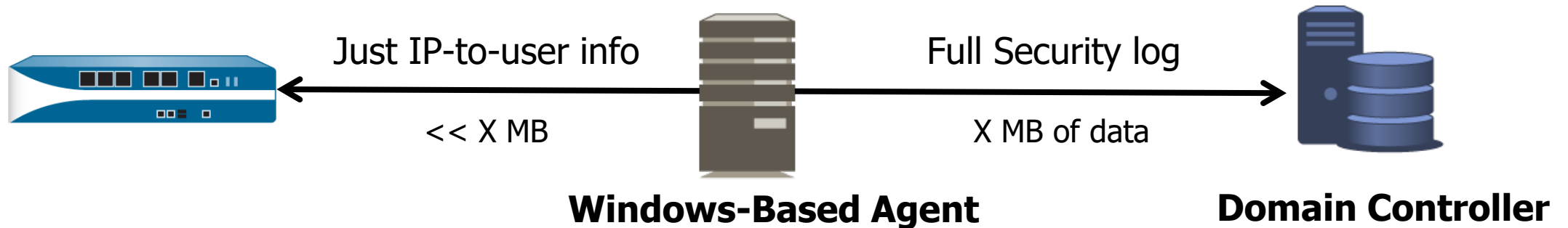
TECHNICAL COLLEGE

User-ID Components

Component	Characteristics
Palo Alto Networks firewall	• Maps IP addresses to usernames • Maps usernames to group names
PAN-OS integrated User-ID agent	• Runs on the firewall • Collects IP address-to-username information
Windows-based User-ID agent	• Runs on a domain member • Collects IP address-to-username information • Sends information to the firewall
Palo Alto Networks Terminal Services agent	• Runs on Microsoft and Citrix terminal servers • Collects IP and port number-to-username information • Sends information to firewall

Integrated Agent Versus Windows-Based Agent

- An integrated agent uses network bandwidth more efficiently.
- For remote sites:
 - Use an integrated agent at the local site, or
 - Install a Windows-based agent at the site.



User-ID overview



User mapping methods overview

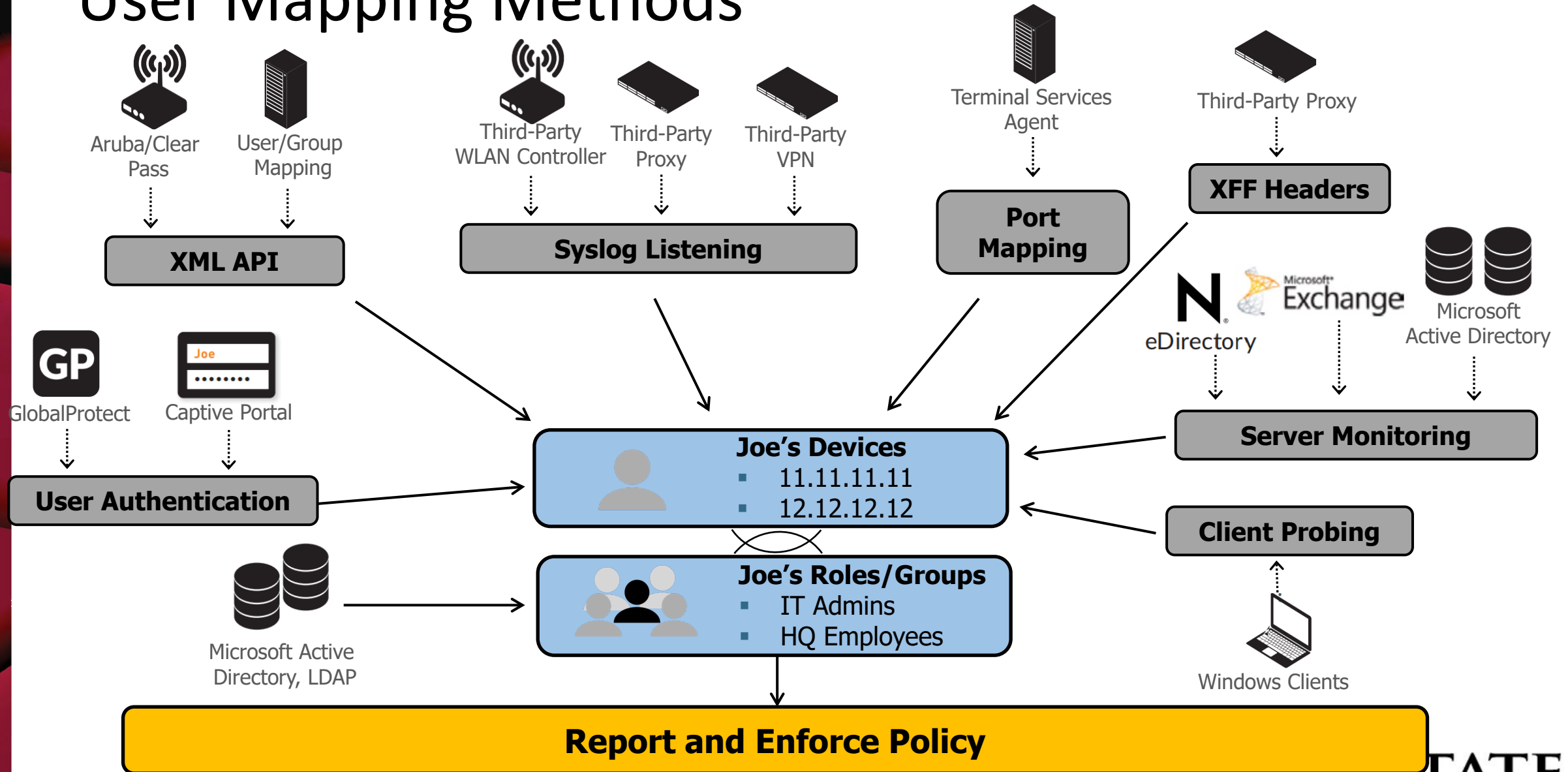
Configure User-ID

PAN-OS integrated agent configuration

Configure group mapping

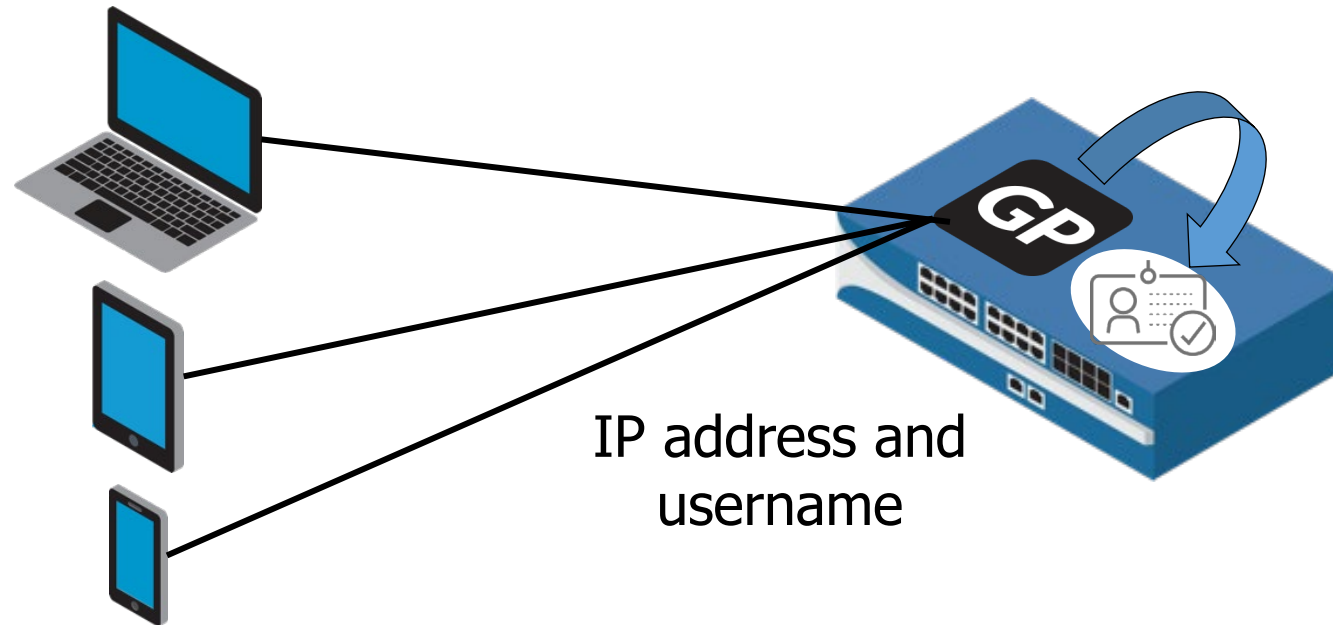
User-ID and Security policy

User Mapping Methods



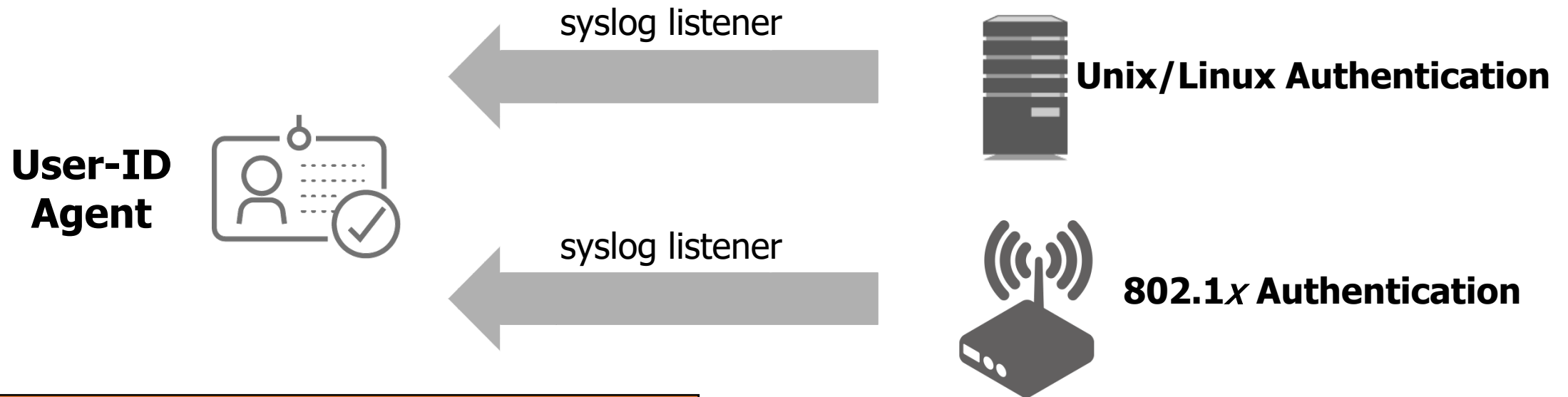
User Mapping Using GlobalProtect

- Every GlobalProtect user is required to enter login credentials to access the firewall.
- GlobalProtect directly adds the username to the firewall's User-ID mapping table.
- GlobalProtect is the best solution for high-security environments.



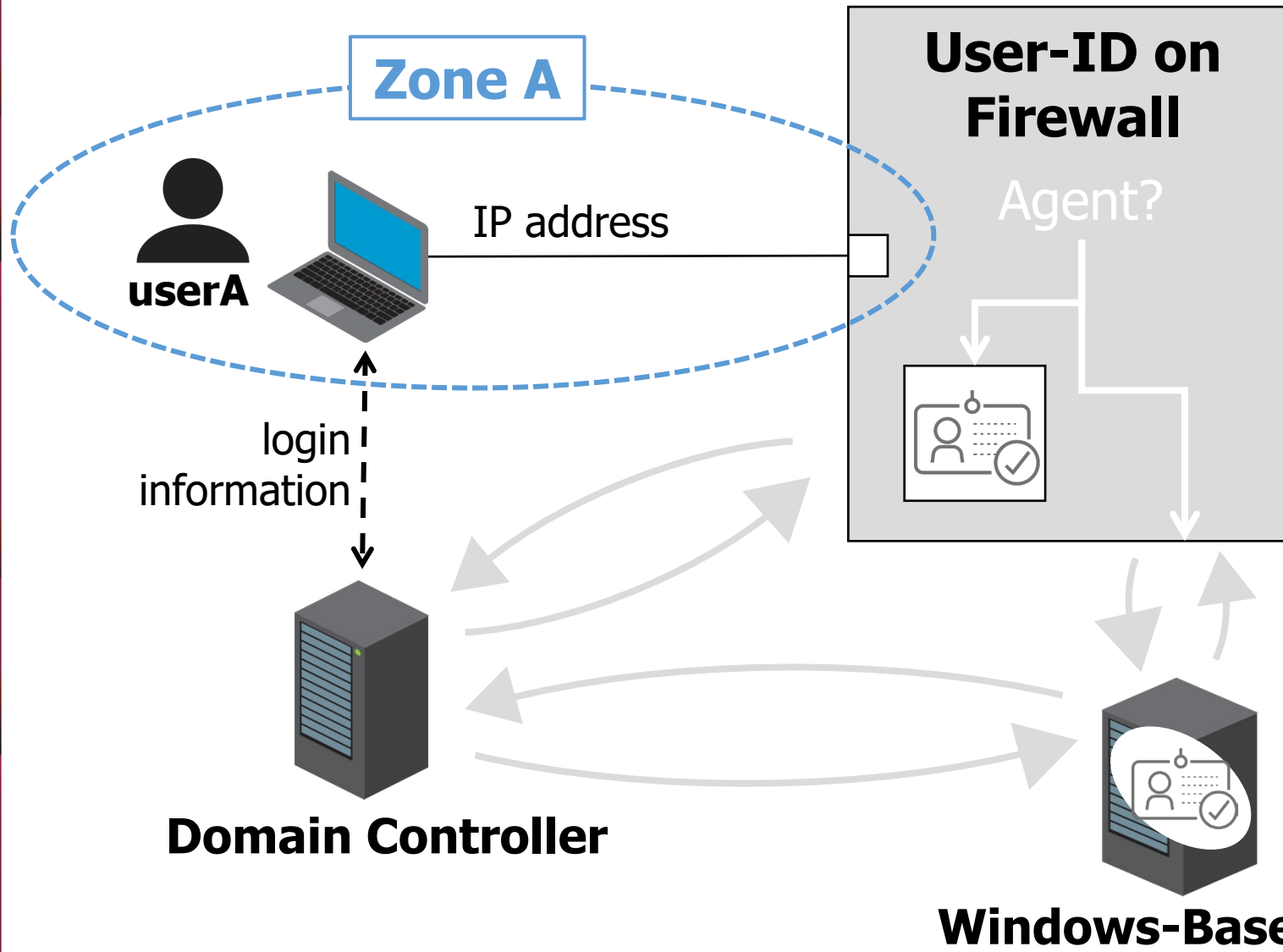
User-ID Syslog Monitoring

- Monitors syslog events for login and logout messages.
- Messages are used to update IP address-to-username mappings.
- Syslog Parse Profiles enable interoperability with diverse syslog types.



For a Tech Doc about this topic, log into Live and search for "Configure User-ID to Monitor Syslog Senders for User Mapping"

User-ID Operation Overview: Domain Controllers

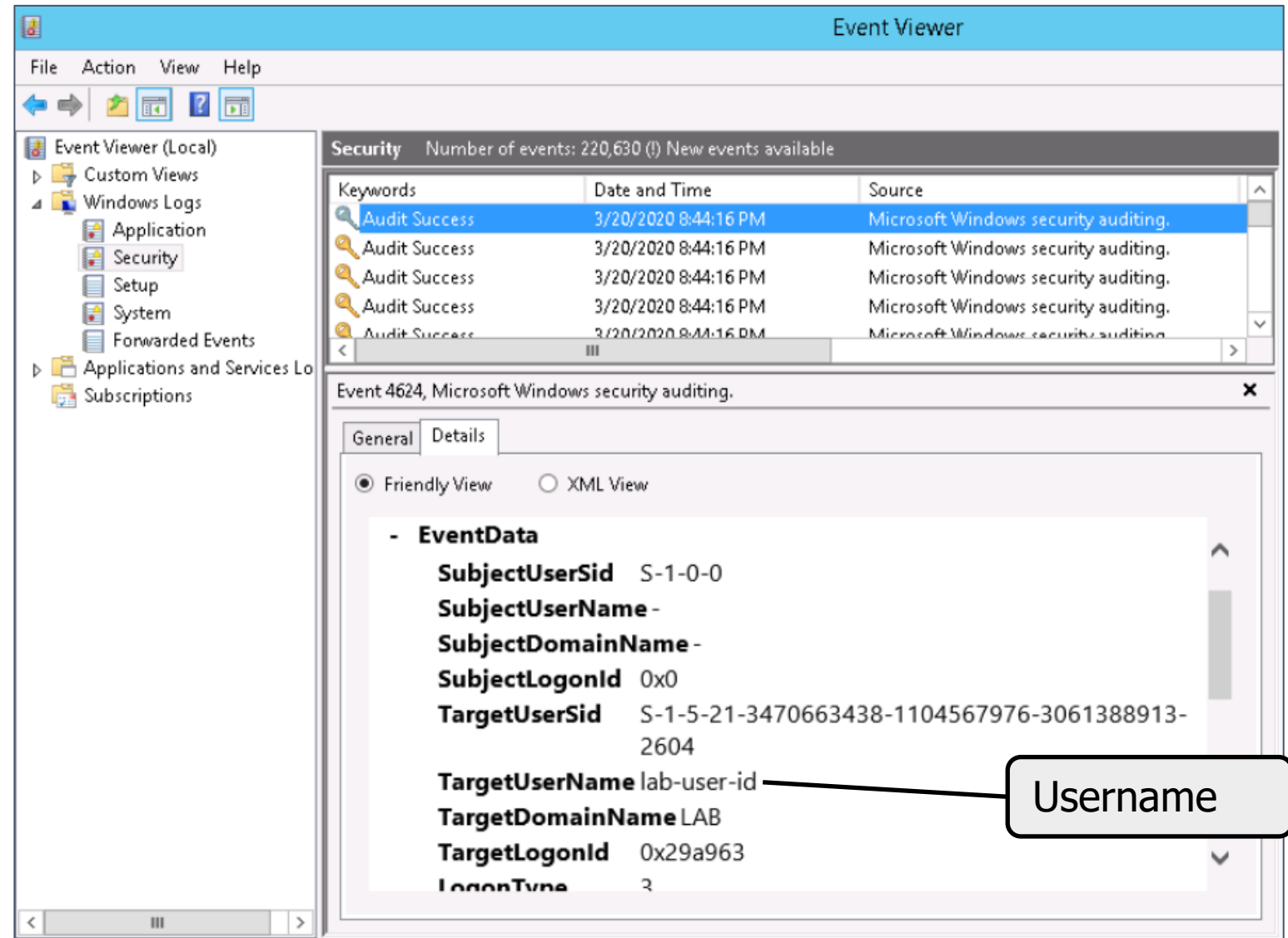


- User-ID enabled on zone?
- Who is agent for domain?
- Query integrated agent for IP/user information, or
- Query Windows-based agent for IP/user information.
- Associate IP with user.
- Associate user with group.
- Check Security policy for match.

Windows-Based Agent

User-ID Domain Controller Monitoring

- Monitors Security logs of domain controllers
- Monitors all domain controllers per domain to get all login and logout events

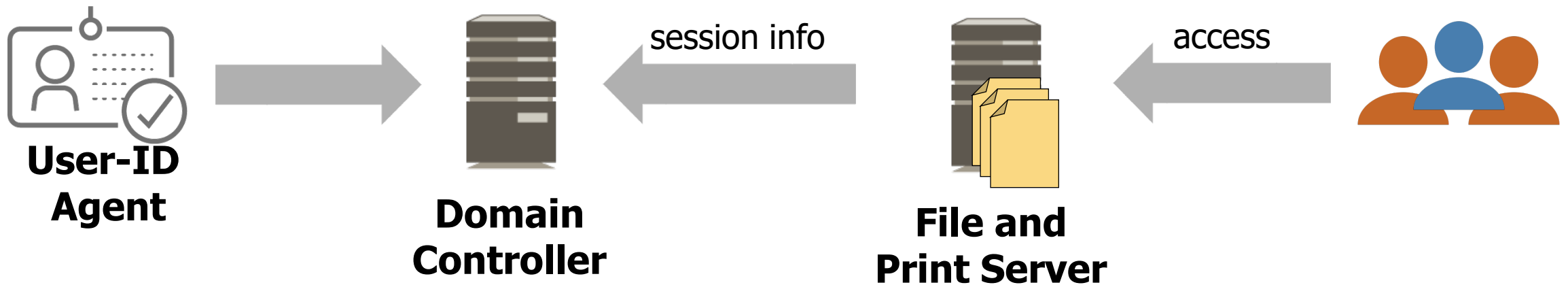


Username

For a Tech Doc about this topic, log into Live and search for "Create a Dedicated Service Account for the User-ID Agent"

User-ID Windows Session Monitoring

- The server logs session information when users connect to shared printers or files.
- Session monitoring is used to maintain known IP address-to-username mappings.



User-ID Mapping Recommendations

If you have ...	Use
GlobalProtect VPN clients	GlobalProtect
Web clients that do not use the domain server	Captive Portal
Non-windows systems, NAC mechanisms such as wireless controllers, 802.1x devices, or proxy servers	syslog listener
Exchange servers, domain controllers, or eDirectory servers	User-ID agent: Server monitoring
Windows file and print shares	User-ID agent: Session monitoring
Multi-user systems such as Microsoft Remote Desktop Services or Citrix Metaframe Presentation Server (XenApp)	Terminal Services agent
Windows clients that often change IP addresses	User-ID agent: Client probing
Devices and applications not integrated with User-ID	XML API

User-ID overview

User mapping methods overview

Configure User-ID



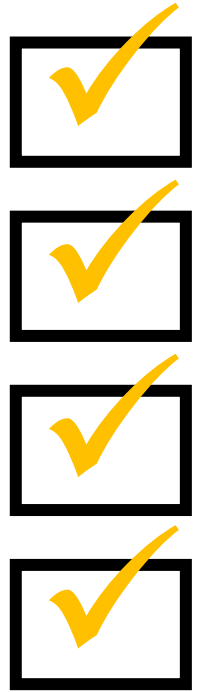
PAN-OS integrated agent configuration

Configure group mapping

User-ID and Security policy

Configure User-ID

1. Enable User-ID by zone.
2. Configure user mapping methods.
3. Configure group mapping (optional).
4. Modify firewall policy rules to use username or group names.



Enable User-ID Per Zone

- Enable User-ID on the source zone where user traffic originates.
- Enable User-ID only for internal zones.
- By default, all subnetworks in the source zone are mapped:
 - Modify using **Include List** or **Exclude List**.

Network > Zones > <select_zone>

Zone

Name

Log Setting

Type

☐ INTERFACES ^

☐ ethernet1/2

User Identification ACL

☒ Enable User Identification

☐ INCLUDE LIST ^

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Device-ID ACL

☐ Enable Device Identification

☐ INCLUDE LIST ^

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

User-ID overview

User mapping methods overview

Configure User-ID

PAN-OS integrated agent configuration



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User-ID and Security policy

Configure the PAN-OS Integrated User-ID Agent

- On the domain controller, create a service account with the required permissions to run the agent.
- On the firewall, define the server's addresses to be monitored.
- Add the service account to monitor the server(s).
- Configure session monitoring.
- Configure WMI probing.
- Commit the configuration and verify agent connection status.



optional

optional



Define the Monitored Server(s)

- Use **Discover** for domain controllers.
- Use **Add** to manually add servers:
 - Required for Exchange, eDirectory, syslog sender

Device > User Identification > User Mapping

Server Monitoring

☐	NAME	ENABLED	TYPE
--------------------------	------	---------	------

User Identification Monitored Server ⓘ

Name: LDAP-Server

Description: LDAP Server Monitoring

☒ Enabled

Type: Microsoft Active Directory

Transport Protocol: WMI

Network Address: ldap.lab.local

Microsoft Active Directory
Microsoft Exchange
Novell eDirectory
Syslog Sender

WMI
WinRM-HTTP
WinRM-HTTPS

Define the User-ID Agent Account

- Necessary permissions are provided if the agent account belongs to:
 - Domain Administrators group, or
 - Server Operators and Event Log Readers groups

Device > User Identification > User Mapping

The screenshot shows the 'Palo Alto Networks User-ID Agent Setup' interface. At the top, there is a navigation bar with tabs: 'User Mapping' (selected), 'Connection Security', 'Terminal Server Agents', 'Group Mapping Settings', 'Trusted Source Address', 'Authentication Portal Settings', and 'Cloud Identity Engine'. Below this, the main title 'Palo Alto Networks User-ID Agent Setup' is displayed with a gear icon to its right. An arrow points from this gear icon to the 'Server Monitor Account' tab. The 'Server Monitor Account' tab is active, showing a form with the following fields:

- Username: lab.local\lab-user-id
- Domain's DNS Name: lab.local
- Password: [masked with dots]
- Confirm Password: [masked with dots]
- Kerberos Server Profile: LDAP-Kerberos (dropdown menu)

Optional Session Monitoring

Device > User Identification > User Mapping

User Mapping | Connection Security | Terminal Server Agents | Group Mapping Settings | Captive Portal Settings

Palo Alto Networks User-ID Agent Setup 

Domain's DNS Name lab.local

Palo Alto Networks User-ID Agent Setup 

Server Monitor Account | **Server Monitor** | Client Probing | Cache | Syslog Filters | Ignore User List

Windows Server Monitoring

☒ Enable Security Log 

Server Log Monitor Frequency (sec)

☐ Enable Session 

Server Session Read Frequency (sec)

Novell eDirectory Monitoring

Server monitoring is enabled by default.

Click to enable session monitoring.

Optional WMI Client Probing

Device > User Identification > User Mapping

User Mapping | Connection Security | Terminal Server Agents | Group Mapping Settings | Captive Portal Settings

Palo Alto Networks User-ID Agent Setup 

Domain's DNS Name lab.local

Kerberos Server Profile LDAP-Kerberos

Palo Alto Networks User-ID Agent Setup 

Server Monitor Account | Server Monitor | **Client Probing** | Cache | Syslog Filters | Ignore User List

☐ Enable Probing

Probe Interval (min)

Integrated agent supports only WMI probing.

Verify Connection Status

Device > User Identification

[User Mapping](#) | [Connection Security](#) | [Terminal Server Agents](#) | [Group Mapping Settings](#) | [Captive Portal Settings](#)

Palo Alto Networks User-ID Agent Setup

Domain's DNS Name

lab.local

Kerberos Server Profile

Enable Security Log

☒

Server Log Monitor Frequency (sec)

2

Enable Session

☐

Server Session Read Frequency (sec)

10

Novell eDirectory Query Interval (sec)

30

Syslog Service Profile

Enable Probing

☐

Probe Interval (min)

20

Enable User Identification Timeout

☐

User Identification Timeout (min)

45

Allow matching usernames without domains

☐

Server Monitoring

☐	NAME	ENABLED	TYPE	NETWORK ADDRESS	STATUS
☐	LDAP-Server	☒	Microsoft Active Directory	192.168.1.20	Connected

User-ID overview

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Configure User-ID

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User-ID and Security policy

LDAP Server Profile

Device > Server Profiles > LDAP > Add

LDAP Server Profile

Profile Name

☐ Administrator Use Only

Server List

NAME	LDAP SERVER	PORT
LDAP-Server1	192.168.1.20	389
LDAP-Server2	192.168.1.21	389

+ Add - Delete

Enter the IP address or FQDN of the LDAP server

Server Settings

Type

Base DN

Bind DN

Password

Confirm Password

Bind Timeout

Search Timeout

Retry Interval

☐ Require SSL/TLS secured connection

☐ Verify Server Certificate for SSL sessions

Where and how to search the LDAP directory tree

Where to connect

active-directory

e-directory

sun

other

Create User-ID Group Mapping Filters

Device > User Identification > Group Mapping Settings > Add

Group Mapping ?

Name

Server Profile | User and Group Attributes | Group Include List | Custom Group

Server Profile Update Interval

Domain Setting

User Domain

Group Objects

Search Filter

Object Class

User Objects

Search Filter

Object Class

☒ Enabled

☐ Fetch list of managed devices

Select LDAP Server Profile.

Dynamically populated based on LDAP server type

Create User-ID Group Mapping Filters (Con't)

Device > User Identification > Group Mapping Settings > Add

Group Mapping

NameLDAP-Group-Mappings

Server Profile | **User and Group Attributes** | Group Include List | Custom Group

User Attributes

NAME	DIRECTORY ATTRIBUTE
Primary Username	sAMAccountName
E-Mail	mail
Alternate Username 1	userPrincipalName
Alternate Username 2	
Alternate Username 3	

Specify **Primary Username** attribute.

Specify up to three alternate attributes.

Group Attributes

NAME	DIRECTORY ATTRIBUTE
Group Name	name
Group Member	member
E-Mail	mail

Filter Groups Sent to the Firewall

Device > User Identification > Group Mapping Settings > Add

Group Mapping

Name

Server Profile | User and Group Attributes | **Group Include List** | Custom Group

Available Groups

→ ×

- cn=enterprise admins
- cn=enterprise read-only domain controll
- cn=group policy creator owners
- cn=lab users**
- cn=protected users
- cn=ras and ias servers
- cn=read-only domain controllers
- cn=schema admins
- cn=winrmremotewmiusers__

Included Groups

- lab\lab users



- Only **Included Groups** are available on drop-down lists in policy rules.
- Shorter lists simplify firewall policy rule administration.

Custom Groups Based on LDAP Filters

Device > User Identification > Group Mapping Settings > Add

Group Mapping

Name LDAP-Group-Mappings

Server Profile | User and Group Attributes | Group Include List | Custom Group

1 item

☐	NAME	LDAP FILTER
☐	Marketing_Grp	(department=Marketing)

+ Add

- Delete

Clone

- Define custom LDAP filters that select group members.
- Assign a custom filter a group name.
- Use a group name in policy rules.

User-ID overview

User mapping methods overview

Configure User-ID

PAN-OS integrated agent configuration

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User-ID and Security policy



Select Users and Groups for a Security Policy

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Users_to_Extranet	Users_Net	universal	Users_Net	any	lab\lab users	any	Extranet	any	any	ssl web-browsing	application-default	Allow
2	Danger-Simulated-Traffic	Danger	universal	Danger	any	lab\lab users	any	Danger	any	any	any	application-default	Allow

Security Policy Rule

General

Source

Destination

Application

Service/URL Category

Actions

Usage

☐ Any	☒ Any	
☒ SOURCE_ZONE ^	☐ SOURCE_ADDRESS ^	select
☐ Users_Net		☐ SOURCE_USER ^
		☒ lab\lab users
+ Add - Delete	+ Add - Delete	+ Add - Delete

- Source user options:
 - any
 - pre-logon
 - known-user
 - unknown
 - select

For a Tech Doc about this topic, log into Live and search for "Building Blocks in a Security Policy Rule"

Dynamic User Groups (DUGs)

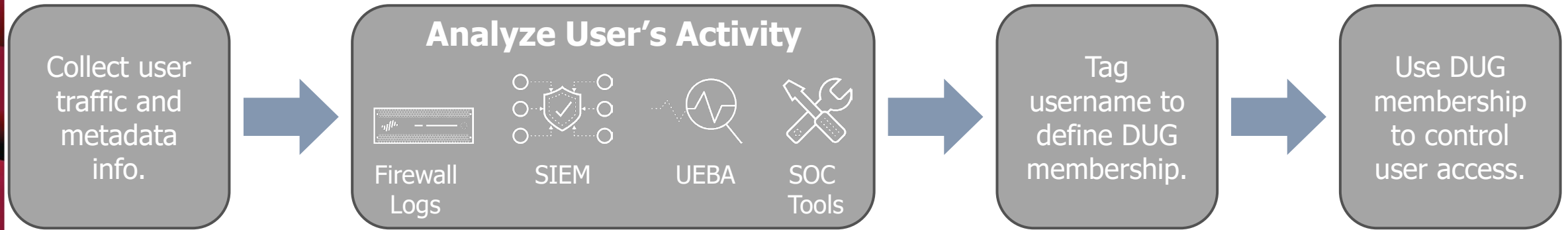
- DUGs control user access to resources managed by firewall policies:
 - Security policy, Authentication policy, Decryption policy, etc.
- User membership in a DUG is dynamic:
 - Only tagged usernames become members of the group.
 - Changes to group membership do not require a commit.

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE			
1	Allow-Temp-Workers	Users_Net	universal	Users_Net	any	Temp-Workers	any	Extranet	any	any	ssl web-browsing	application-default	Allow

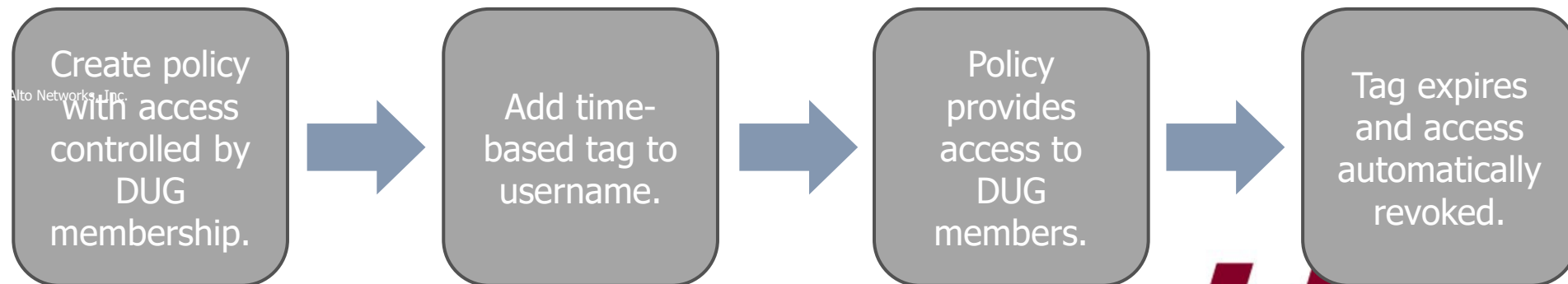
Only tagged
users are
members.

Two Example Use Cases

Leverage User's Entire Known Security State:



Use Time-Based User Access Controls:



Module Summary

Now that you have completed this module, you should be able to:

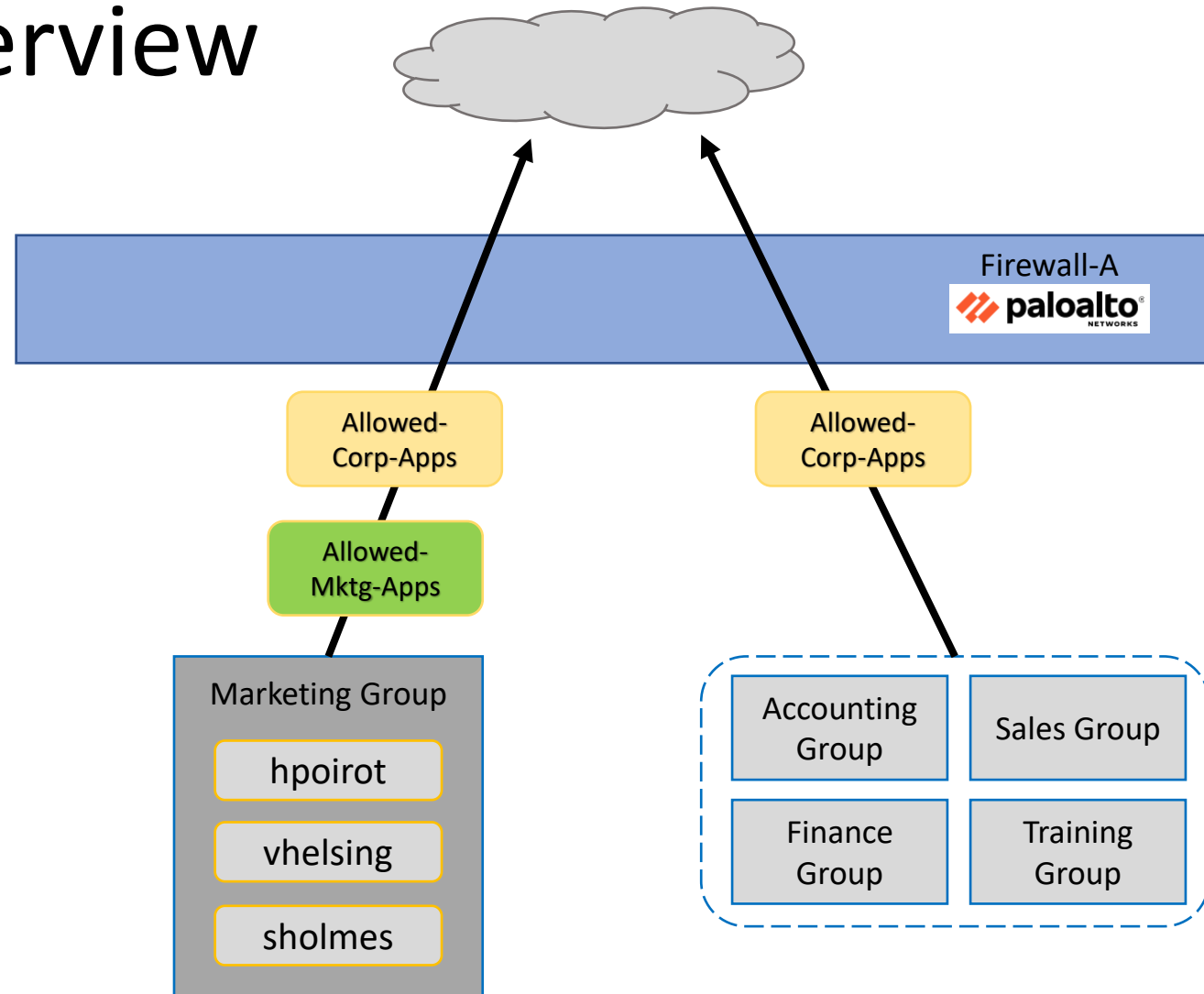
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Questions



Lab 11: Overview



Lab 11: Controlling Access to Network Resources with User-ID

- Examine current configuration
- Enable User-ID technology on the Acquisition zone.
- Generate traffic
- Modify Security policy to meet requirements

End Content

Unit 12 Homework

- Always check Blackboard for assignments and when they are due.
- Unit 12 Lab: Netlabs Lab 11 – Controlling Access to Network Resources with User-ID
- Unit 11 DoS Protection Video
- Unit 12 Quiz
- Units 8 – 12 Quiz 3